



ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2011

BIOLOGY
9648/01

HIGHER 2
2011

26 August

Paper 1 Multiple Choice

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, pencil clips, highlighters, glue or correction fluid.

Write your name, Centre number and index number on the Answer Sheet provided.

There are **forty** questions in this paper. Answer **all** questions. For each question there are four possible answers, **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate answer sheet.

Read the instructions on the Answer Sheet very carefully.

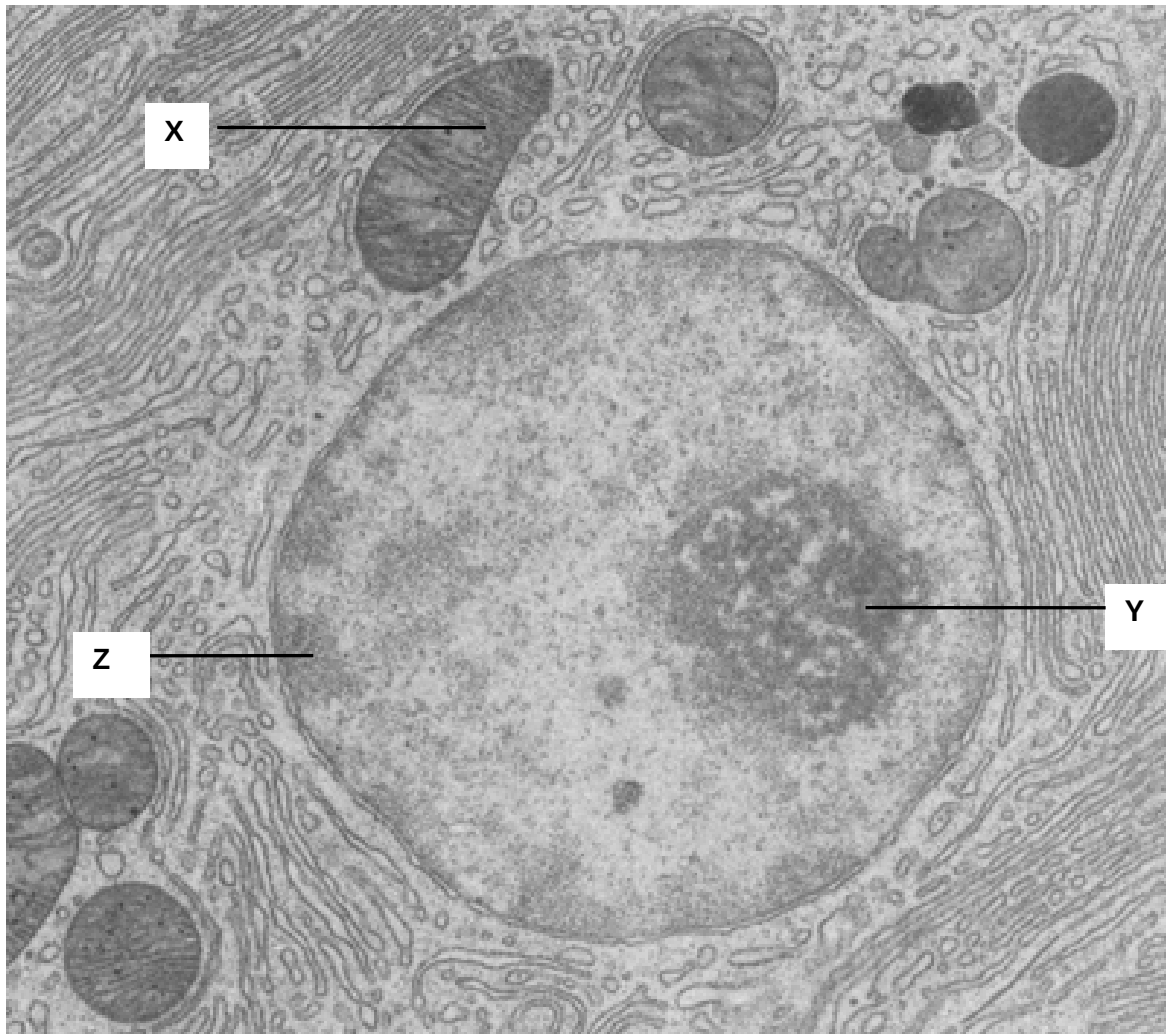
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Calculators may be used.

This document consists of **21** printed pages.

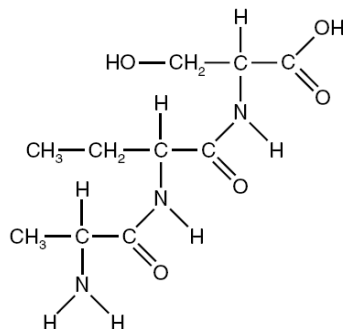
1 The electronmicrograph below shows the structures found in a cell.



Which of the following statements is true for structures X, Y and Z?

	X	Y	Z
A	Contains reduced NAD^+ and reduced FAD	Transcription of gene coding for ribosomal RNA	Contains heterochromatin which is transcriptionally inactive
B	Involved in oxidative phosphorylation	Transcription of gene coding for ribosomal protein	Contains heterochromatin which is transcriptionally active
C	Contains reduced NAD^+ and reduced FAD	Involved in assembly of ribosomal subunits	Contains euchromatin which is transcriptionally inactive
D	Involved in oxidative phosphorylation	Involved in synthesis of ribosomal subunits	Contains euchromatin which is transcriptionally active

- 2 The diagram below shows a biological molecule found in the living cells.



The following are some statements concerning the molecule shown above.

- 1 It is soluble in water.
- 2 Double bonds between the carbon and oxygen atoms increase membrane fluidity.
- 3 It can act as a buffer against extreme pH change.
- 4 It can undergo condensation to form a polymer.

Which of the above statements are true about this molecule?

- A** 1 and 3
B 1 and 4
C 2 and 3
D 2 and 4

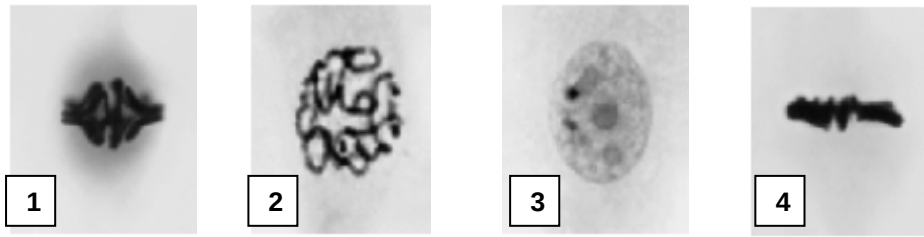
- 3 Creatine kinase is active in the intermembrane space of mitochondria. The enzyme activity shows a reading of 50 arbitrary units (A.U.) under normal cellular condition. The enzyme activity of creatine kinase was also measured after incubation in the following solutions:

- 1 Deionized water.
- 2 Acetic acid.
- 3 Sodium hydroxide.

Which of the following findings are consistent with the expected results?

	1	2	3
A	30 A.U.	50 A.U.	10 A.U.
B	50 A.U.	30 A.U.	10 A.U.
C	30 A.U.	10 A.U.	50 A.U.
D	10 A.U.	30 A.U.	50 A.U.

- 4 The photomicrographs below depict the different stages of a typical mammalian cell cycle.



Which of the above photomicrograph(s) could be obtained from the following cells?

- | | Lymphocyte | Erythrocyte |
|----------|-------------------|--------------------|
| A | All of the above | 3 only |
| B | 1, 2 and 4 | None of the above |
| C | All of the above | None of the above |
| D | 3 only | 1, 2 and 4 |
- 5 Meiosis is known to produce four daughter cells containing half the original number of chromosomes in the parent cell.

The table below shows the number of chromosomes in each gamete after meiosis has taken place in human germ cells.

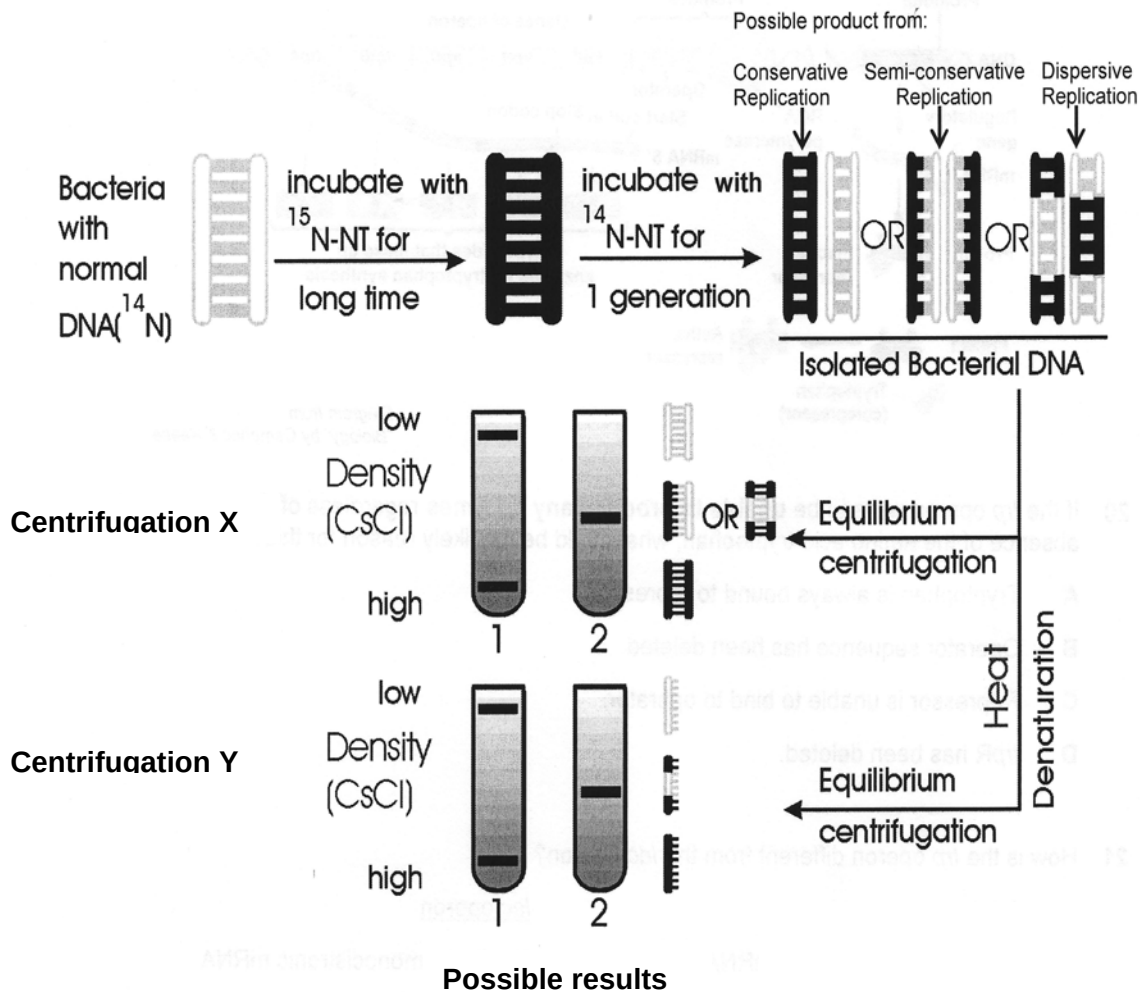
Gametes 1-4 were derived from human germ cell X, whereas gametes 5-8 were derived from human germ cell Y.

Human germ cell X		Human germ cell Y	
Gametes	Number of chromosomes	Gametes	Number of chromosomes
1	22	5	23
2	24	6	23
3	24	7	22
4	22	8	24

Which of the following description is consistent with the data shown in the table above?

- A** Homologous chromosomes could not pair up to form bivalents in germ cell X.
- B** A pair of chromatids could not separate during anaphase II in germ cell X.
- C** Centromere of a pair of chromatids failed to divide during anaphase II in germ cell Y.
- D** Homologous chromosomes could not separate during anaphase I in germ cell Y.

- 6 The Meselson-Stahl experiment is shown below. The experiment was done to show that DNA replication occurs by a semi-conservative mode rather than a conservative or dispersive mode.



Given that replication is semi-conservative, which result (tube 1 or 2) would you expect to see for centrifugation X and Y respectively? In addition, which of the other modes of replication would the expected results disprove?

	Centrifugation X		Centrifugation Y	
	Correct result	Disproves	Correct result	Disproves
A	1	Dispersive	1	Conservative
B	2	Conservative	2	Dispersive
C	1	Dispersive	2	Conservative
D	2	Conservative	1	Dispersive

- 7 The following coding sequence of DNA is taken randomly from a bacterial genome.

3' TTACGCTTCGAAATAGGAATATCATAGGCT 5'

This sequence is cloned into a plasmid and transformed into a suitable host. What would be the first four amino acids of a polypeptide generated from this sequence as expressed by the host?

The mRNA codons for some amino acids are shown in the table below:

Arg	CGA, CGG, AGA, AGG	Leu	CUU, CUC, CUA, CUG
Asp	GAU, GAC	Lys	AAA, AAG
Ile	AUU, AUC, AUA	Phe	UUU, UUC
Start	AUG	Ser	UCA, UCG, AGU, AGC
Stop	UAG, UGA, UAA	Tyr	UAU, UAC

- A Met-Arg-Ser-Phe
 B Met-Arg-Ser-Lys
 C Met-Ile-Phe-Leu
 D Met-Tyr-Lys-Asp
- 8 Listed below are some antibacterial drugs that can affect the synthesis of bacterial proteins.

Anti-microbial drug	Rifampicin	Streptomycin	Tetracycline
Mode of action	Binds to RNA polymerase	mRNA misread during translation	Prevents binding of tRNA to ribosome

Which row shows the correct effects of these drugs on bacterial protein synthesis?

	Rifampicin	Streptomycin	Tetracycline
A	Defective protein synthesised	mRNA does not bind to ribosome	Polymerization of amino acids does not occur
B	mRNA not synthesised	Defective protein synthesised	Polymerization of amino acids does not occur
C	mRNA not synthesised	mRNA does not bind to ribosome	Transcription prevented
D	Transcription prevented	Defective protein synthesised	mRNA does not bind to ribosome

- 9 What do chromosomal aberrations and gene mutations have in common and how are they different?

	Similarity	Difference
A	Both may involve addition of nucleotides.	Gene mutations always produce dominant alleles but not chromosomal aberrations.
B	Both may not result in disorders.	Gene mutations do not involve inversions but inversions of segments of chromosomes do occur.
C	Both affect nucleotide sequence in DNA molecule.	Gene mutations occur within a chromosome but chromosomal aberrations may occur across chromosomes.
D	Both may not result in a difference in protein expression.	Chromosomal aberrations are more harmful than gene mutations.

- 10 How do viruses cause diseases in animals?

- 1 They disrupt normal cell functions of the host cell such as protein synthesis.
- 2 They disrupt the oncogenes of the host cell causing uncontrolled cell division.
- 3 Their viral proteins and glycoproteins on the surface membrane of host cells cause these cells to be recognized and destroyed by the body's immune defenses.
- 4 They deplete the host cell of cellular material essential for metabolic functions.

- A 1, 2 and 3
 B 1, 3 and 4
 C 2, 3 and 4
 D All of the above

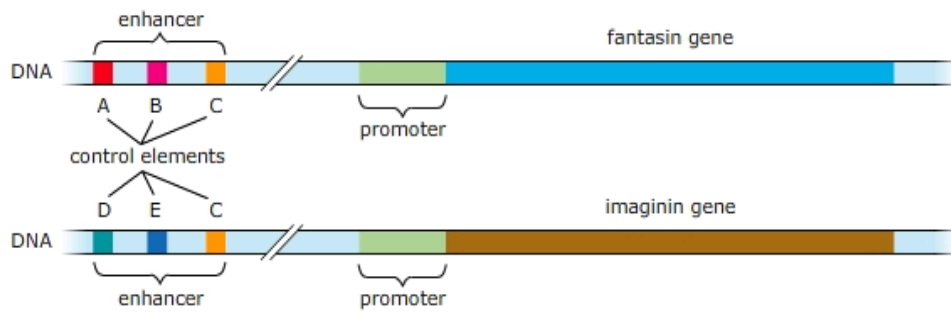
- 11 Hepatitis C virus (HCV) can only replicate by injecting its genome into human cells upon infection. HCV is a positive-sense RNA virus, i.e. its genomic material is RNA that can be used for translation immediately. Which of the following are most likely **not** needed for HCV replication?

- 1 DNA-dependent RNA polymerase
- 2 Reverse transcriptase
- 3 Ribosomes
- 4 RNA-dependent RNA polymerase

- A 1 and 2
 B 1 and 3
 C 1, 2 and 4
 D 1, 3 and 4

- 12** A bacterial cell, infected by a lambda phage, gets co-infected with another lambda phage at a later point in time. This did not result in the death of the bacterial cell. Which of the following statements is a/are possible reason/s for the phenomenon?
- 1 The presence of the first lambda phage inhibits the reproductive cycle of the second lambda phage.
 - 2 The transcription and translation of viral DNA is not initiated.
 - 3 A repressor protein is synthesized which blocks the expression of phage genes.
 - 4 Mutation of host genome results in the inability of lambda phage to enter host cell.
- A** 1 only
B 1 and 2
C 2 and 3
D 3 and 4
- 13** Which of the following mutations will bring about active transcription of the *trp* operon in *E. coli* ?
- A** A mutation in the repressor gene that strengthens the affinity of the repressor for the co-repressor.
- B** A mutation in the repressor gene that strengthens the affinity of the repressor for the inducer.
- C** A mutation in the repressor gene that weakens the affinity of the repressor for the operator.
- D** A mutation in the repressor gene that weakens the affinity of the repressor for the inducer.

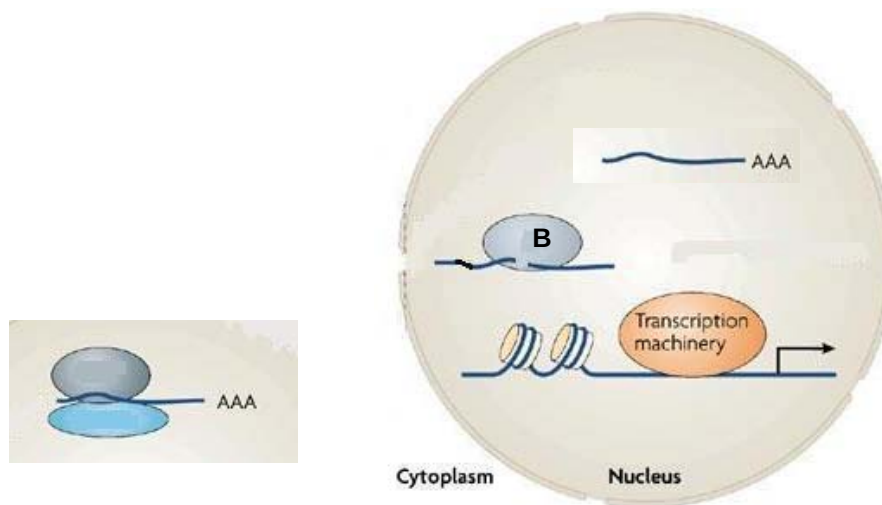
- 14 The diagram below shows the control elements and two genes found in the human genome.



Adapted from *Biology* by Campbell and Reece © 2008 Pearson Education, Inc.

Which of the following statements about the above genes is true?

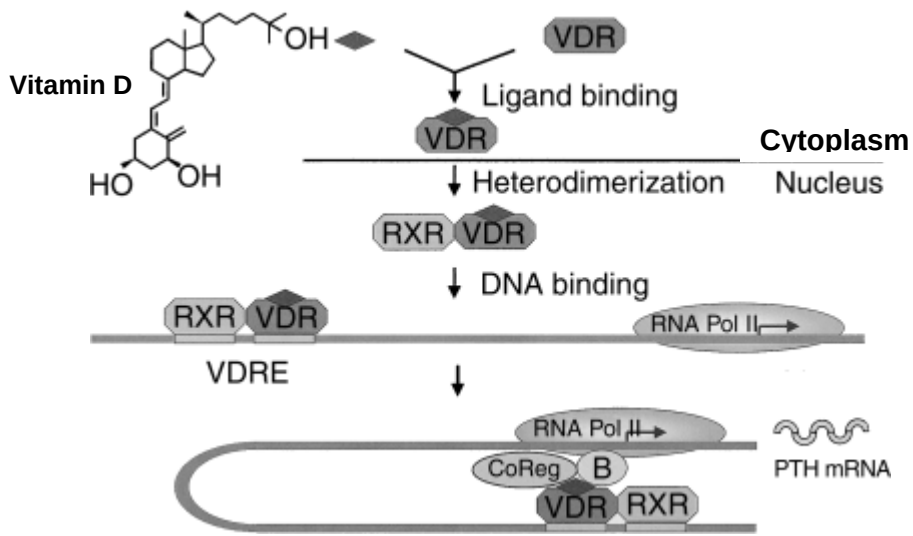
- A Control elements A, B, and C are proximal control elements for the fantasin gene.
 - B The fantasin gene and the imaginin gene have identical enhancers.
 - C The fantasin gene will be transcribed at a high level when activators specific for control elements A, B, and C are present in the cell.
 - D The imaginin gene will be transcribed at a high level when repressors specific for the imaginin gene are present in the cell.
- 15 The figure below shows the process of gene expression in a eukaryotic cell.



A gene mutation resulted in a non-functional molecule **B**. What would be the product of the gene expression?

- A A different polypeptide was produced.
- B A variety of polypeptides was produced.
- C No polypeptide will be produced.
- D An mRNA that will be degraded immediately by endonucleases in the cytoplasm.

- 16** Vitamin D helps the body to increase its calcium concentration for the production of strong bones. The figure below illustrates the effect of Vitamin D binding to its receptor (VDR).



The parathyroid hormone (PTH) increases calcium concentration in the body by increasing calcium absorption in the intestine and by increasing calcium reabsorption in the kidney.

Which of the following statements is/are true?

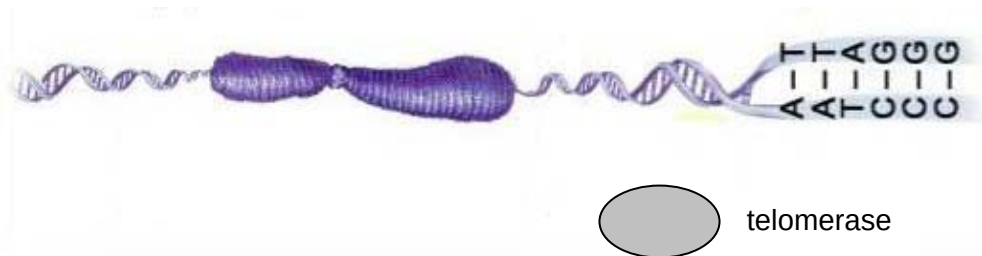
- 1 VDR is an intracellular receptor for Vitamin D.
- 2 RXR-VDR-Vitamin D complex functions as an enhancer and brings about an increase in the expression of PTH.
- 3 Vitamin D brings about an increase in the transcription of PTH gene.

- A** 1 only
B 1 and 3 only
C 2 and 3 only
D 1, 2 and 3

- 17** Which combination of factors will affect the translational control of gene expression?

	Presence of carbohydrate group attached to polypeptide	Presence of repressor bound to 5' UTR of mRNA	Presence of ubiquitin attached to polypeptide
A	Yes	No	Yes
B	Yes	Yes	No
C	No	Yes	No
D	No	No	No

- 18** Ras protein is a G-protein involved in a cell-signalling pathway. This protein is coded for by a proto-oncogene. Which of the following would lead to the cell undergoing uncontrolled cell division?
- A** Duplication of the Ras promoter that controls the proto-oncogene.
 - B** Formation of multiple copies of the Ras proto-oncogene.
 - C** Point mutation within the proto-oncogene, forming mutant Ras which is constitutively active.
 - D** Translocation of the proto-oncogene upstream of a hyperactive promoter.
- 19** Telomeres are non-coding DNA sequences usually found at the ends of chromosomes. In certain cells, telomere length can be increased by the action of the telomerase enzyme, coded for by the *TERC* gene.



Which of the following statements is/are **false**?

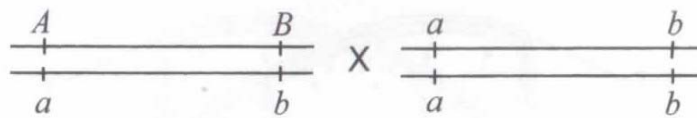
- 1 Telomerase is active only in stem cells and germ cells.
- 2 Telomere lengths shorten with every cell division due to incomplete replication of the lagging strand of DNA.
- 3 Telomerase adds nucleotide sequences to the 5' end of the DNA.
- 4 A gene mutation to *TERC* leads to reduced telomere length.

- A** 1 only
- B** 1 and 2 only
- C** 1 and 3 only
- D** 3 and 4 only

- 20 In pigeons, the alleles of the gene controlling eye colour are codominant and are carried on the X chromosome. When a black-eyed female was mated with a white-eyed male, the F_1 generation consisted of black-eyed males and grey-eyed females.

What phenotypic ratio would be expected in the F_2 generation?

- A 1 black-eyed male: 1 white-eyed male: 2 grey-eyed females
 B 1 black-eyed male: 1 white-eyed male: 1 grey-eyed female: 1 black-eyed female
 C 1 black-eyed male: 1 grey-eyed male: 1 white-eyed female: 1 black-eyed female
 D 2 black-eyed males: 1 grey-eyed female: 1 white-eyed female
- 21 A cross between individuals of genotypes **AaBb** and **aabb** is carried out as follows:



The genotype and number of progeny obtained are as shown below:

Genotype	Number of Progeny
$\frac{A}{a} \frac{B}{b}$	40
$\frac{a}{a} \frac{B}{b}$	
$\frac{a}{a} \frac{B}{b}$	11
$\frac{A}{a} \frac{b}{b}$	
$\frac{A}{a} \frac{b}{b}$	9
$\frac{a}{a} \frac{B}{b}$	
$\frac{a}{a} \frac{b}{b}$	40
$\frac{a}{a} \frac{b}{b}$	

What is the map distance, in map units, between the two genes?

- A 9
 B 11
 C 20
 D 40

- 22 In certain breeds of mice, the two allelic pairs, **C/c** and **A/a** are known to regulate the formation of coat colour. Coloured coat is produced in the breed carrying at least one copy of **C** allele and two copies of **a** alleles, whereas agouti coat is produced when the breed carries at least one copy of **C** and **A** alleles. Albino coat is derived from the breed homozygous for **c** allele.

Which of the following description is valid when two agouti mice heterozygous for both **C** and **A** genes are crossed?

- A** The expected ratio of agouti mice to coloured mice and to albino mice is 9:6:1.
- B** The phenotypic effect coded by the allele **A** is masked in the mice homozygous for allele **c**.
- C** The production of coat colour is regulated sequentially first by gene **A** and then gene **C**.
- D** Agouti is an intermediate coat colour resulting from the codominant effect between gene **A** and gene **C**.
- 23 In a comparative study of brinjal plants, a test cross was made between the variety of plant producing purple and long brinjal and the variety producing green and short brinjal. The results of the following F_1 generation are shown below:

Phenotypes	Number
Purple, Long	28
Purple, Short	30
Green, Long	26
Green, Short	34
Total number	118

$$\chi^2 \text{ test} \quad \chi^2 = \sum \frac{(O - E)^2}{E} \quad \nu = c - 1$$

key to symbols

Σ = 'sum of ...'
 ν = degrees of freedom
 c = number of classes
 O = observed 'value'
 E = expected 'value'

degrees of freedom	probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

Which of the following statements is **false**?

- A** The probability that the difference between observed and expected values is due to chance is greater than 10%.
- B** The two genes coding for the colour and shape are not linked.
- C** The difference between the expected number and the observed number of the phenotypes occurred by chance.
- D** The calculated X^2 value is greater than the critical X^2 value.

- 24** It is suggested by scientists that prosopagnosia, a condition in which an individual is unable to recognise or distinguish between the faces of other humans, was caused by a single dominant gene. In preliminary studies, up to 2% of tested individuals were affected.

Which of the following would support the hypothesis that it is caused by a single, dominant gene?

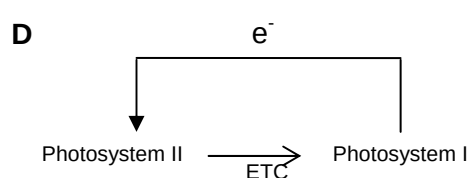
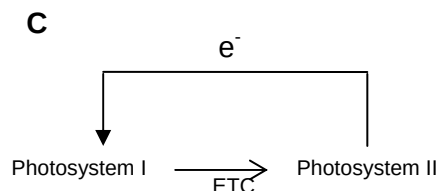
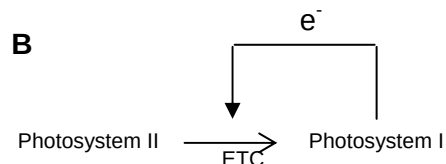
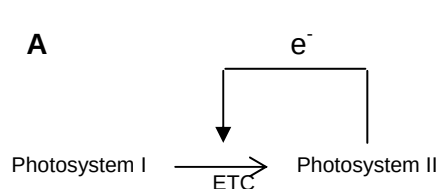
- A** Only males are prosopagnosics.
 - B** As long as one of the parents is a prosopagnosic, all the offspring will be prosopagnosic.
 - C** No offspring of two normal individuals are prosopagnosic.
 - D** The severity of the trait in individuals follows a normal distribution.
- 25** European rabbits, homozygous for black fur, were crossed with rabbits homozygous for white fur. The F_2 generation was obtained from inbreeding of F_1 generation and raised at three different temperatures.

The table shows the results in the F_2 generation.

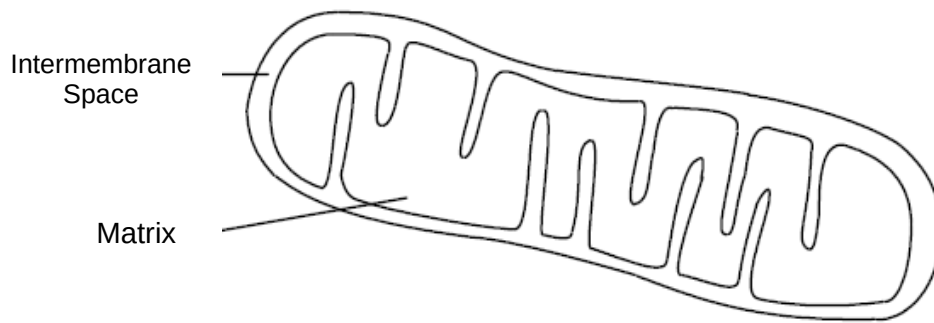
Temperature/ $^{\circ}\text{C}$	Result
4	All white
15	3/4 grey, 1/4 white
28	3/4 black, 1/4 white

Which statement **best** explains these results?

- A** White fur and black fur illustrate codominance at 15°C .
 - B** Heterozygous rabbits have white fur only at 4°C .
 - C** White fur is dominant and expressed at all temperatures.
 - D** Black fur is dominant at higher temperatures but recessive at 4°C .
- 26.** Which diagram correctly represents the process of cyclic photophosphorylation?

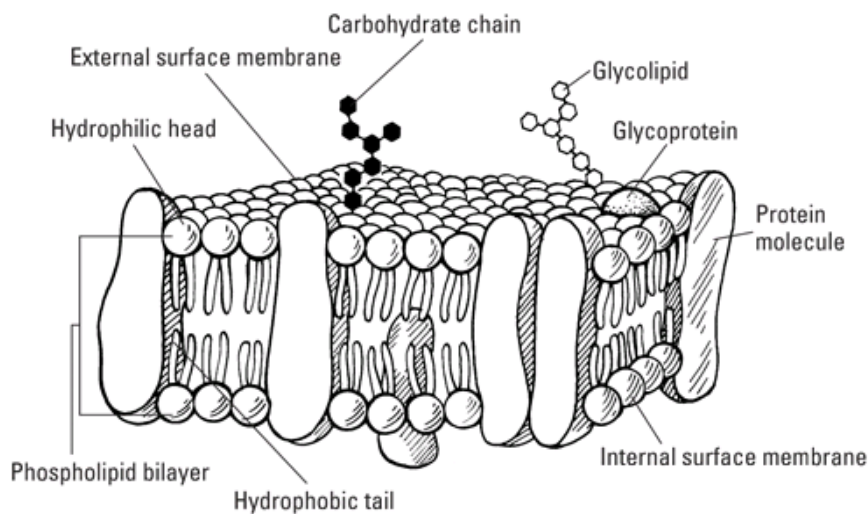


- 27 The diagram below shows a mitochondrion as seen under the electron microscope.



What is the advantage of having a small volume inside the inter-membrane space of the mitochondrion?

- A A high electron concentration is rapidly developed.
 - B A high proton concentration is rapidly developed.
 - C Enzymes of the Krebs cycle are highly concentrated.
 - D Protein electron carriers are highly concentrated.
28. The figure below shows a cross-section of a typical cell membrane.



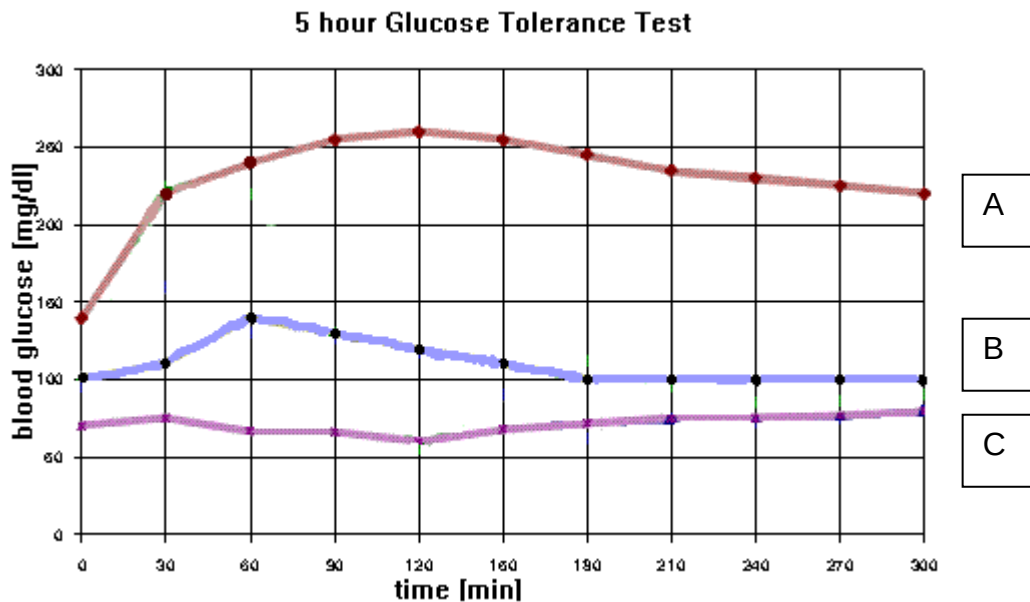
Which of the following statements about the fluid mosaic model of cell surface membranes is **false**?

- A 'Fluid' refers to the phospholipids and proteins being free to move within the membrane.
- B 'Mosaic' refers to the scattered manner in which membrane proteins are embedded in the cell membrane.
- C Phospholipids can move laterally and even "flip-flop" within the cell membrane.
- D Phospholipids and proteins are held together by hydrogen bonds.

- 29 During synaptic transmission, which substance enters and which substance leaves the presynaptic knob?

	Substance entering presynaptic knob	Substance leaving presynaptic knob
A	calcium ions (Ca^{2+})	neurotransmitter
B	chloride ions (Cl^-)	neurotransmitter
C	neurotransmitter	calcium ions (Ca^{2+})
D	neurotransmitter	potassium ions (K^+)

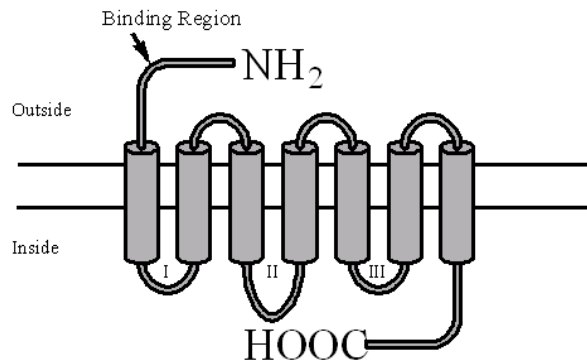
- 30 The 5-hour glucose tolerance test involves drinking a 100g glucose solution, and then measuring the blood glucose level over 5 hours. Persons **A**, **B** and **C** took the test, and their results are shown in the figure below.



Which of the following can be correctly concluded from the graph?

- A In a normal healthy individual, it takes three hours for insulin to restore normal blood glucose levels.
- B Person **A** suffers from Type 2 adult-onset diabetes.
- C The beta cells of the islets of Langerhans in person **A** are non-functional.
- D The pancreas released insulin 60 minutes after it had detected the rise in blood glucose level.

- 31** The figure below shows a protein receptor on a cell membrane.

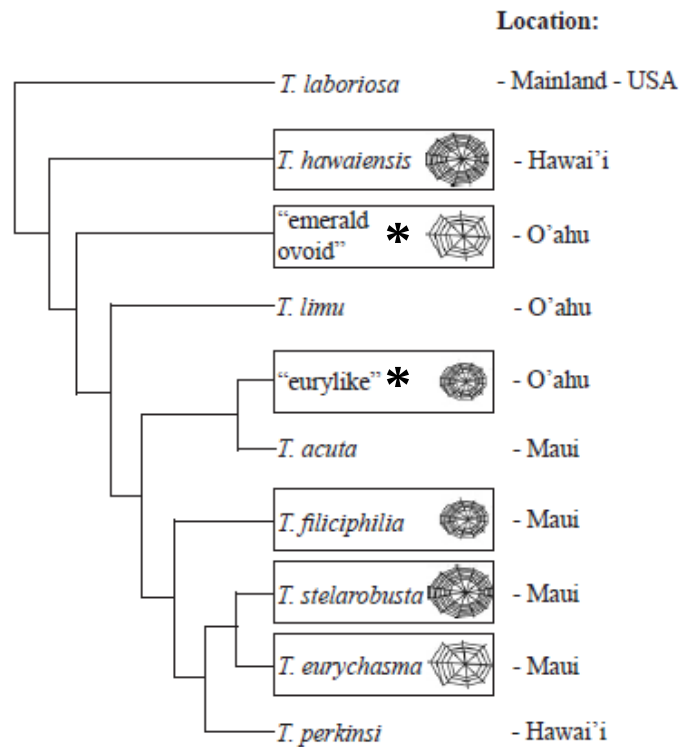


Which statement correctly describes how this receptor responds upon binding to a ligand?

- A** It enters the nucleus, undergoes a conformational change and binds to DNA.
 - B** It undergoes a conformational change, enters the nucleus and binds to DNA.
 - C** It undergoes conformational change and binds to G-protein.
 - D** It undergoes dimerisation, and hence a conformational change.
- 32** Which of the following statements about the neutral theory of molecular evolution is/are true?
- 1 Neutral mutation can only occur in non-coding regions.
 - 2 Neutral mutations do not come under selection pressure.
 - 3 Neutral mutations may be fixed in the population over time, due to genetic drift.
 - 4 Neutral theory is the basis for the molecular clock concept.

- A** 2 only
- B** 2 and 4 only
- C** 2, 3 and 4 only
- D** All of the above

- 33 The cladogram below shows the classification of a group of spiders found on the Hawaiian islands. An asterisk (*) indicates that this species of spider has yet to be assigned a scientific name.



Based on information from the diagram above, deduce which of the following spiders are the most closely related species.

- A *T. filiciphilia* and "eurylike"
 - B *T. hawaiiensis* and "emerald ovoid"
 - C *T. stelarobusta* and *T. eurychasma*
 - D *T. stelarobusta* and *T. filiciphilia*
- 34 What is the primary reason for natural selection working on every natural population?
- A The genetic variability among organisms of different species.
 - B The innate tendency of organisms in populations to compete against one another.
 - C The need for organisms to fight against natural disasters and diseases.
 - D The tendency for populations to produce more offspring than can be supported.

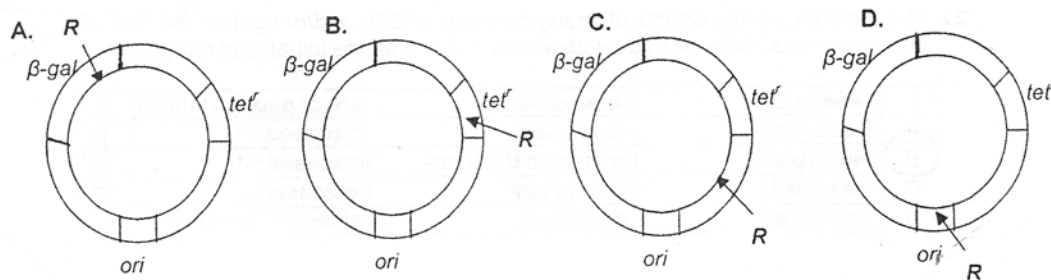
- 35** Calcitonin is a protein hormone found in humans, fish, birds, and mammals. It is 32 amino acids long. The table below shows the amino acid sequence of calcitonin in various organisms and the number of amino acid differences when compared with human calcitonin.

Organism	Amino acid sequence of calcitonin	Number of amino acid differences
Human	CGNLSTCMLGTYTQDFNKFHTFPQTAIGVGAP-NH ₂	-
Salmon	CSNLSTCVLGKLSQELHKLQTYPRTNTGSGTP-NH ₂	16
Eel	CSNLSTCVLGKLSQELHKLQTYPRTDVGAGTP-NH ₂	16
Rat	CGNLSTCMLGTYTQDLNKFHTFPQTSIGVGAP-NH ₂	2
Chicken	CASLSTCVLGKLSQELHKLQTYPRTDVGAGTP-NH ₂	17

What conclusion can be drawn from the data above?

- A** Humans and salmon are more closely related than salmon and eel.
B Humans are most closely related to rats.
C Salmon and chicken share a recent common ancestor.
D Salmon and eel are more closely related than salmon and chicken.
- 36** A synthetic plasmid having an origin of replication (*ori*), a restriction site (*R*), two genetic markers, β -galactosidase gene (*β -gal*) and tetracycline-resistance gene (*tet^r*), was used as a vector to transfer the human growth hormone gene into bacteria. The *β -gal* gene codes for an enzyme β -galactosidase, which breaks down a colourless compound called X-gal to a blue compound.

At which position should the restriction site (*R*) be located on the plasmid in order to identify desired transformed bacteria on an agar medium containing both X-gal and tetracycline?



- 37** Each of the statements below is concerned with the success or failure of gene therapy as a permanent cure for genetic disorders.
- 1 Somatic cells cannot pass the modified gene on to any offspring.
 - 2 The treatment does not last long as the treated cells die and are replaced.
 - 3 Post-translational modification may be missing in the treated cells.
 - 4 The treatment does not result in all cells having the therapeutic gene.

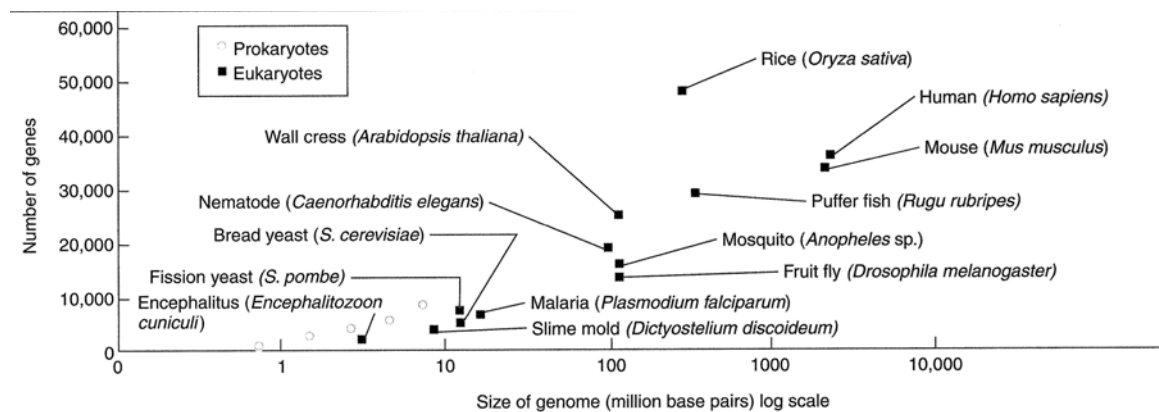
Which statement/s explain/s the limited success of gene therapy as a permanent cure for genetic disorders in human populations?

- A** 1 only **B** 1 and 3 **C** 1, 2 and 3 **D** 1, 2 and 4

- 38 Anti-thrombin (a protein involved in blood clotting) and human growth hormone are examples of human proteins that can be produced via the genetic engineering technique.

How would the process which produces anti-thrombin differ from the one that produces bacterial antibiotic resistance protein if both were to be carried out in bacteria?

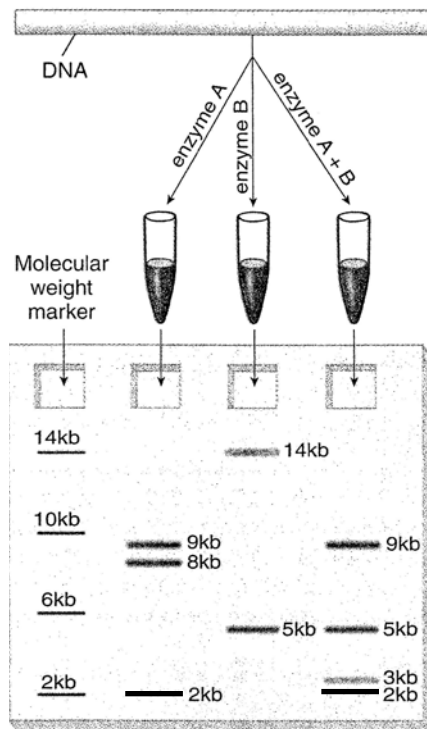
- A Antibiotic resistance would not be used as a selection marker for successfully transformed bacteria for the process that produces the antibiotic resistance protein.
 - B The anti-thrombin gene insert would require an additional eukaryotic promoter.
 - C Anti-thrombin would require additional post-translational modification.
 - D Different plasmids would have to be used.
- 39 The diagram below shows the number of genes and genome size of various types of organisms.



Which of the following conclusions **cannot** be drawn from the diagram above?

- A Generally, eukaryotic genomes are larger and have more genes than prokaryotic genomes.
- B Humans have about twice as many genes as *Drosophila* and fewer genes than rice.
- C The mouse genome is nearly as large as the human genome.
- D The complexity of an organism is reflected in the number of genes it contains.

- 40** Digestion of a 19 kb DNA molecule with restriction enzyme A yields 3 fragments: 9 kb, 8 kb and 2 kb. Digestion of the same molecule with restriction enzyme B yields fragments: 14 kb and 5 kb. Finally, digestion with both restriction enzymes, A and B together, yields fragments: 9 kb, 5 kb, 3 kb and 2 kb, as shown in the diagram below.



Which of the following shows the correct restriction map indicating the positions of the restriction enzyme A and B cleavage sites?

- A**
- | | | | |
|-----|-----|-----|-----|
| 9kb | 3kb | 5kb | 2kb |
| | A | B | A |
- B**
- | | | | |
|-----|-----|-----|-----|
| 3kb | 5kb | 2kb | 9kb |
| B | A | A | |
- C**
- | | | | |
|-----|-----|-----|-----|
| 5kb | 3kb | 9kb | 2kb |
| A | A | | B |
- D**
- | | | | |
|-----|-----|-----|-----|
| 2kb | 3kb | 5kb | 9kb |
| A | B | A | |

End of Paper